

Sisipho Zinja

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I am a machine learning researcher with expertise in biomedical AI applications and large-scale distributed systems. I have developed predictive learning models for cancer evolution and treatment resistance at UCL Cancer Institute, and hold an MSc in Artificial Intelligence with Distinction. My research interests include ethical deep learning for oncology, computational biology, and distributed machine learning systems. I have previous experience scaling ML systems for 300M+ users at Twitter, with a focus on distributed training optimization and recommendation algorithms, and I am eager to deepen my empirical AI safety research skills.

Education

MSc Artificial Intelligence City St George's, University of London London, UK <i>Distinction · Focus: Machine Learning, Deep Learning, Multi-Agent Systems</i>	Aug 2024 – Jan 2026
BSc Computer Science University of Rochester New York, USA <i>Focus: Data Structures & Algorithms, AI, Computational Systems · Renaissance & Global Scholarship · Bank of America Women in Computing Scholarship</i>	Aug 2016 – May 2020
Semester Abroad – Game Development & Big Data Danish Institute for Study Abroad Copenhagen, Denmark	Jan 2019 – May 2019

Employment History

Research Engineer • MSc Dissertation	Dec 2024 – Present
- Designed and implemented a multimodal deep learning framework in Python that integrated high-dimensional clinical data with non-Euclidean tree topologies using Tree-LSTM networks and a custom Cox partial likelihood loss. - Led end-to-end experimentation on tumor phylogeny models, turning an open-ended research question into concrete model architectures, training regimes, and evaluation protocols. - Collaborated with clinical experts in a human-in-the-loop validation setup to label and audit tumor phylogenetic indices, aligning model behavior with real-world biological ground truth. - Built robust data cleaning and labeling workflows for the TRACERx cohort, transforming raw, noisy genomic data into high-fidelity training sets that improved survival prediction C-index from 0.70 to 0.83 (+18%). - Debugged complex failures across data preprocessing, model training, and evaluation, iteratively refining the research pipeline under uncertainty.	
Research Software Engineer • University College London Cancer Institute	Jan 2025 – Oct 2025
- Developed and deployed cloud-integrated machine learning pipelines in Python that were used by 50+ researchers to model cancer evolution and treatment resistance, contributing to an 85% improvement in therapy identification. - Owned the full lifecycle of research pipelines—from prototype models to productionized workflows—so that domain scientists could run large-scale experiments without engineering bottlenecks. - Refactored and documented legacy gene sequence-to-expression modeling workflows, reducing engineering onboarding time by 30% and improving reproducibility for empirical studies. - Streamlined data labeling and evaluation pipelines for high-dimensional cancer genomic datasets, improving data integrity and supporting reliable, data-driven clinical analyses.	
Software Engineer • Twitter/X	Sept 2020 – Jul 2024
- Conducted research and engineering on distributed training using TensorFlow model and data parallelism, achieving up to 100x efficiency on sparse recommendation models powering feeds for 300M+ users. - Collaborated with applied researchers and engineers to design, run, and analyze large-scale training experiments, iterating on model configurations and infrastructure to improve throughput and cost efficiency. - Scaled backend distributed systems for the authentication platform, improving monitoring and reducing latency across billions of daily requests under stringent reliability constraints.	
Software Engineer Intern • Twitter/X	May 2019 – Aug 2019
- Improved global traffic routing systems handling billions of daily requests, enhancing reliability and reducing latency worldwide. - Led development of an internal tool that allowed developers to route test requests directly to backend services bypassing the Twitter Front End, while maintaining high quality standards and data protection requirements.	

Skills

• Python	• TensorFlow	• Generative Adversarial Networks (GANs)	• Computer Vision
• ML	• Keras	• Distributed Training	• Human-in-the-loop (HITL) workflows
• Deep Learning	• Reinforcement Learning	• Distributed Systems	• Data Labeling
• PyTorch	• LLM Fine-tuning (RLHF)	• Multi-modal Data Fusion	• Synthetic Data Generation